

AI Critique Report

During the performance testing design and implementation phases, several recurring patterns of limitations and inaccuracies were observed in the AI's behavior. The most prominent issue was the AI's lack of situational awareness regarding file storage conventions. It frequently defaulted to saving executable scripts and data files into the documentation directories rather than the dedicated execution environments, requiring continuous manual corrections to ensure the automated k6 pipelines could locate their dependencies.

Furthermore, the AI exhibited a significant flaw in assessing hardware constraints during the workload modeling phase. Rather than basing the test capacity on the Docker container limits where the system under test was actually deployed, it incorrectly relied on the host machine's total resources. This misidentification led to initial proposals of target virtual users that were highly unrealistic and would have prematurely overwhelmed the micro-tier architecture, undermining the integrity of the performance baseline.

The AI also demonstrated a tendency toward blind compliance over technical verification. For instance, when requested to generate environment configuration files, it immediately complied without verifying whether the testing framework natively supported them. This resulted in the creation of redundant files that had to be manually deleted later. Additionally, the AI frequently hallucinated technical capabilities, particularly regarding data cleanup. It repeatedly proposed using direct SQL queries—which violates black-box testing principles—or assumed the existence of non-existent API endpoints, failing to acknowledge that a manual database reset was the only viable teardown method.

Finally, the AI showed poor judgment regarding test data isolation. It attempted to reuse the same seed data across completely different testing scenarios, specifically carrying over data from the read-heavy test into the endurance test. This cross-contamination could have mutated the shared state and invalidated the long-term soak test results. Consequently, explicit instructions had to be provided to force the AI to generate isolated data seeding scripts for each distinct test suite.